COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010	**SECTION:**	CPE21S4
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Seatwork 7.2 **Algorithm Analysis: Best Case and Worst Case Time Complexity**

1. PROCEDURE OUTPUT

1. Finding an Element

```
None
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Answer:

Best Case: T(n) = 1 → O(1)

Worst Case: T(n) = 1 + n → O(n)

Explanation : Found in First and Last Element

2. Printing All Elements

```
None
for (int i = 0; i < n; i++)
{
    cout << arr[i] << endl;
}
```

Answer:

Best Case: T(n) = 1 +n → O(n)

Worst Case:T(n) = → O(n)

Explanation: It prints every element, so there is no early stopping.

3. Nested Loops

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

Answer:

Best Case: T(n) = (n) (n) = n^2 → O(n^2)

Worst Case: O(n^2)

Explanation: the loops always run the same

4. Conditional Processing

```
None
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
else
{
```

```
    cout << "Small input";
}
```

Answer:

Best Case: T(n) = 1 → O(1)

Worst Case: T(n) = 1 + n O(n)

Explanation: it either performs one action or processes all n elements

5. Fixed Inner Loop

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
```

Answer:

Best Case: T(n) = (n) (10) = 10n → O(n)

Worst Case: O(n)

Explanation: the inner loop runs a fixed 10 times for each of the n iterations

6. Two Consecutive Loops

```
None
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}
```

```
for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

Answer:

Best Case: T(n) = 1 + n + n = 1 + 2n →O(n)

Worst Case: O(n)

Explanation: two loops run n times each, giving 2n

7. Dividing by Two

```
None
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

Answer:

Best Case: T(n) → O(1)

Worst Case: T(n) → O(log n)

Explanation: The value is repeatedly divided by two

8. Nested Loop With Different Limits

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
```

Answer

Best Case:T(n) = → $O(n^2)$

Worst Case:T(n) = → $O(n^2)$

Explanation: the nested loops procedure about n^2 / 2 operation

9. Early Termination

```
None
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
```

Answer:

Best Case: T(n) = → O(1)

Worst Case: T(n) = → O(n)

Explanation: the search can stop immediately or check every element

10. Two-Level Search

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
    }
}
```

Answer:

Best Case: T(n) = → O(n)

Worst Case: T(n) = → $O(n^2)$

Explanation: the inner loop may stop early run fully for every outer iteration

11. Linear Loop With Logarithmic Inner Work

```
None
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Answer:

Best Case: T(n) = O (n log n)

Worst Case: T(n) = O (n log n)

Explanation: n iterations each perform log n work

12. Conditional Nested Loop

```
None
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

Answer

Best Case: T(n) = O(1)

Worst Case: T(n) = $O(n^2)$

Explanation:: it either finds the target immediately or executes two nested loops

13. Finding the Maximum

```
None
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Answer:

Best Case: T(n) = O(n)

Worst Case: T(n) = O(n)

Explanation: every element must be checked to find the maximum

14. Three Nested Loops

```
None
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Answer:
Best Case: $T(n) = O(n^3)$

Worst Case: $T(n) = O(n^3)$

Explanation: three nested loops each run n times

15. Comprehensive Analysis

```
None
for (int i = 0; i < n; i++)
```

Answer:

Best Case: $T(n) = O(1)$

Worst Case:$T(n) = O (n \log n)$

Explanation: it may return immediately or perform an iteration with log n work each

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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